

PROJECT DESIGN PHASE

Project Name: Script-Controlled ACL – Restrict Record Access Based on Field Value

Module 2: Proposed Solution

1. Executive Summary

The proposed solution introduces a script-controlled Access Control List (ACL) framework to enforce record-level security in ServiceNow. The solution evaluates user roles together with the Branch field value before allowing access to records. For this project, users with the required **bb1** role can view EEE branch records, while administrators retain full access regardless of branch. Separate CREATE, WRITE, and DELETE ACLs provide controlled permissions for additional record operations.

2. Core Features & System Capabilities

Role-Based Access Control: Uses custom roles such as bb1, bb2, bb3, and bb4 to control access to different record operations.

Field-Based Access Condition: Uses the Branch field with the condition **Branch is EEE** to restrict READ access to the required records.

Script-Controlled Decision: Uses an advanced ACL script to allow administrators full access, allow users with the required role, and deny access for other users.

Operation-Specific ACLs: Provides separate READ, CREATE, WRITE, and DELETE ACLs for controlled record-level permissions.

Custom Institution Details Table: Protects records in the u_institution_details table containing Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, and Description.

3. Proposed Workflow Steps

Step 1: Create the EEE test user and custom roles bb1, bb2, bb3, and bb4.

Step 2: Create the u_institution_details table and configure the required fields.

Step 3: Create records with different Branch values such as ECE, EEE, and CSE.

Step 4: Create the READ ACL with bb1, the **Branch is EEE** data condition, and the script-controlled access logic.

Step 5: Create separate CREATE, WRITE, and DELETE ACLs using bb2, bb3, and bb4.

Step 6: Impersonate the appropriate users and verify that access changes according to the assigned roles and record values.

4. Key Business Outcomes

Security: Prevents unauthorized users from viewing or modifying restricted branch records.

Record-Level Control: Evaluates record field values together with user roles to make access decisions.

Administrative Access: Ensures administrators retain full access to records regardless of branch.

Controlled Operations: Separates READ, CREATE, WRITE, and DELETE permissions to enforce appropriate access.

Data Protection: Protects institution records from unauthorized viewing, modification, creation, and deletion across ServiceNow interfaces. The source document explicitly identifies these READ, WRITE, CREATE, and DELETE ACLs as the project's intended record-level security outcome.

This follows the sample's **Module 2: Proposed Solution → Executive Summary → Core Features → Proposed Workflow → Key Business Outcomes** structure.